Selim Amar

Contact Information

Stanford, California, USA `selama@stanford.edu`

Summary

I am a mathematician with a background in analysis, partial differential equations, and physics.

Education

Stanford University

PhD in Mathematics, (2022-Present); GPA: 4.0
Expected date of graduation: June 2027

McGill University

B.Sc. in Honours Mathematics and Physics, (2019-2022); GPA: 4.0

Research Experience

2022-Present — Doctoral Research on Microlocal Analysis.
PhD Advisor: András Vasy, Department of Mathematics, Stanford University

Developed new analytic methods for non-elliptic PDEs. Proved Fredholm estimates for wave-type operators in the presence of normally hyperbolic trapping. Defined Feynman propagators and computed the index of Lorentzian Dirac operators. Techniques involve spectral theory, functional analysis, pseudodifferential operators and dynamics.

2021 — Peeling theorem in higher dimensions.
Supervisor: N. Kamran, Department of Mathematics, McGill University

Derived asymptotics for waves on higher-dimensional spacetimes using the Newman-Penrose formalism. Techniques involved asymptotic analysis, ODE theory and differential geometry.

2020 — Discrete exterior calculus.
Supervisor: G. Tsogtgerel, Department of Mathematics, McGill University.

Conducted numerical experiments on simulations of PDEs using the discrete exterior calculus and compared the observed convergence rate with the expected one. Techniques involved finite element methods, discretization of manifolds and numerical linear algebra. Coding done with Python.

Publications

S. Amar, *Feynman Propagators on Exponentially Decaying Perturbations of Stationary Spacetimes with Compact Cauchy Surface*, arXiv Preprint (2026).
S. Amar, *Fredholm Theory of Non-Elliptic Operators in the Presence of Normally Hyperbolic Trapping*, arXiv Preprint (2025).
S. Amar, *A Peeling Theorem for the Weyl Tensor in Higher Dimensions*, Classical and Quantum Gravity (2022).

Relevant Talks

Fredholm Theory of Non-Elliptic Operators with Trapping
(McGill University Geometric Analysis Seminar, January 8, 2026)

Fredholm Theory of Non-Elliptic Operators with Trapping
(Berkeley Spectral Theory Seminar, November 5, 2025)

Neural networks and PDEs: from physics-informed learning to unstable solutions
(Stanford University Student Learning Seminar on Mathematics and Computation, May 13, 2026)

Selected Honors and Awards

FRQNT Doctoral Training Scholarship (2024–); Gowrisankaran Prize in Mathematics & Statistics (2022); Horace Watson Prize in Physics (2022); NSERC Undergraduate Research Award (2021).

Relevant Skills

Languages: English, French
Programming: Python — NumPy, SciPy, Matplotlib
Mathematics: PDEs, Microlocal analysis, Wave equations, Finite element methods

Teaching Experience

Calculus Math 21 (Fall 2026; Teaching Assistant)

Calculus Math 21 (Spring 2026; Teaching Assistant)

Calculus Math 21 (Spring 2025; Teaching Assistant)

Differential Equations with Linear Algebra, Fourier Methods, and Modern Applications Math 53 (Spring 2024; Teaching Assistant)

Linear Algebra and Matrix Theory Math 113 (Winter 2024; Course Assistant)

Calculus Math 21 (Spring 2023; Course Assistant)

Applied Matrix Theory Math 104 (Fall 2022; Course Assistant)

Selected Conferences and Workshops Attended

2nd OIST-Oxford-SLMath Summer Graduate School on Analysis and Partial Differential Equations (OIST, June 22-July 3, 2026)

MAMA26: Modern Applications of Microlocal Analysis
(University College London, February 16–20, 2026)

3rd Simons Math Summer Workshop: Partial Differential Equations of Classical Physics
(Simons Center for Geometry and Physics, July 14-25, 2025)

Nonlinear Waves and Relativity
(Erwin Schrödinger International Institute for Mathematics and Physics, June 17-21, 2024)

From Microlocal to Global Analysis, 75th Birthday Conference of Richard Melrose
(MIT, May 10-12, 2024)

Spring School on Soliton Dynamics (Texas A&M University, March 14-16, 2024)

Spectral Theory and Mathematical Relativity
(Erwin Schrödinger International Institute for Mathematics and Physics, June 19-23, 2023)